

Homework 2:

Kernel Density Estimation

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Da un secolo, oltre.

Accenni Teorici

- Formula KDE**

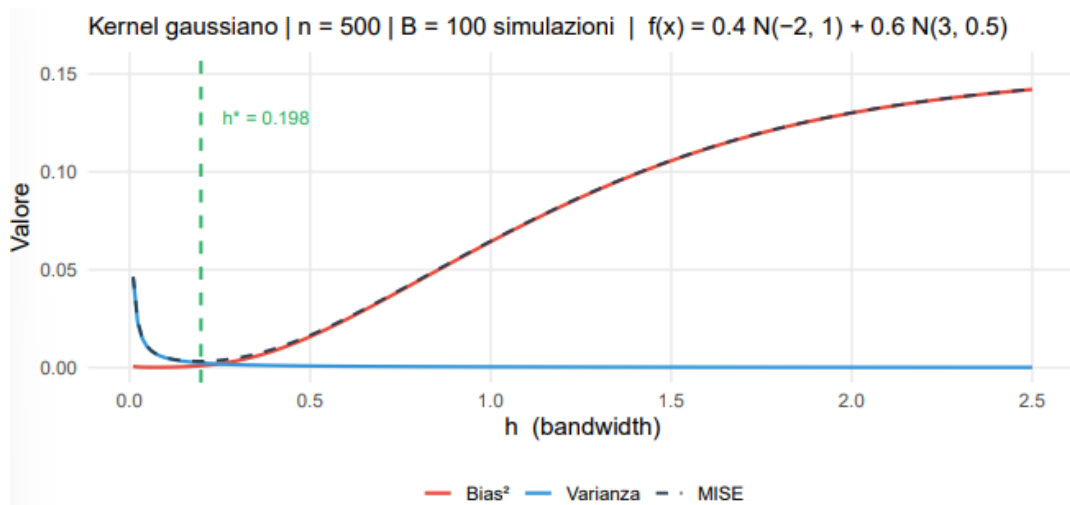
$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

- Kernel Gaussiano**

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$$

$$\text{MISE}(\hat{f}) = \mathbb{E} \left[\int (\hat{f}(x) - f(x))^2 dx \right] = \underbrace{\int \text{Bias}^2[\hat{f}(x)] dx}_{\uparrow \text{ con } h} + \underbrace{\int \text{Var}[\hat{f}(x)] dx}_{\downarrow \text{ con } h}$$

Grafico MISE: Tradeoff *Varianza-Bias*²



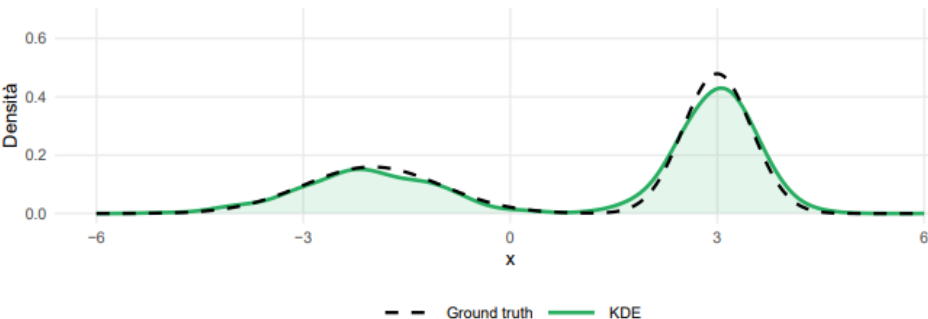
L'importanza della *bandwidth* (h)

Data Generating Process

$$f(x) = 0.4 N(-2, 1) + 0.6 N(3, 0.5)$$

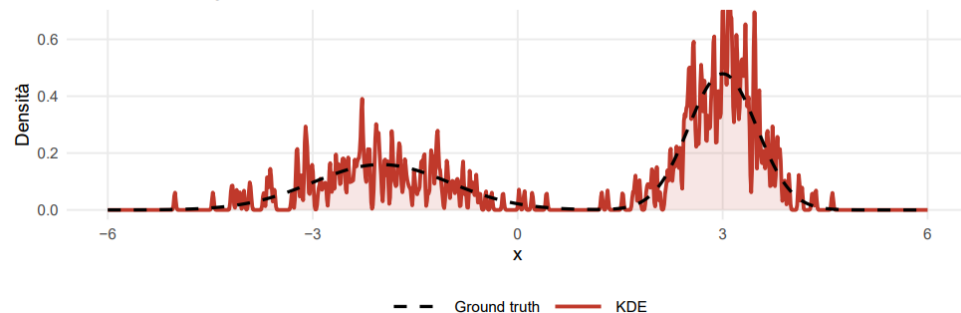
Buona scelta

$h = 0.30 \rightarrow$ buon fit



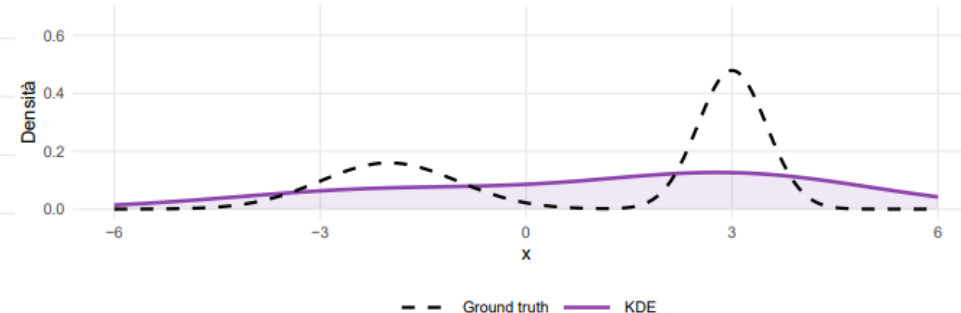
Undersmoothing

$h = 0.01 \rightarrow$ overfitting



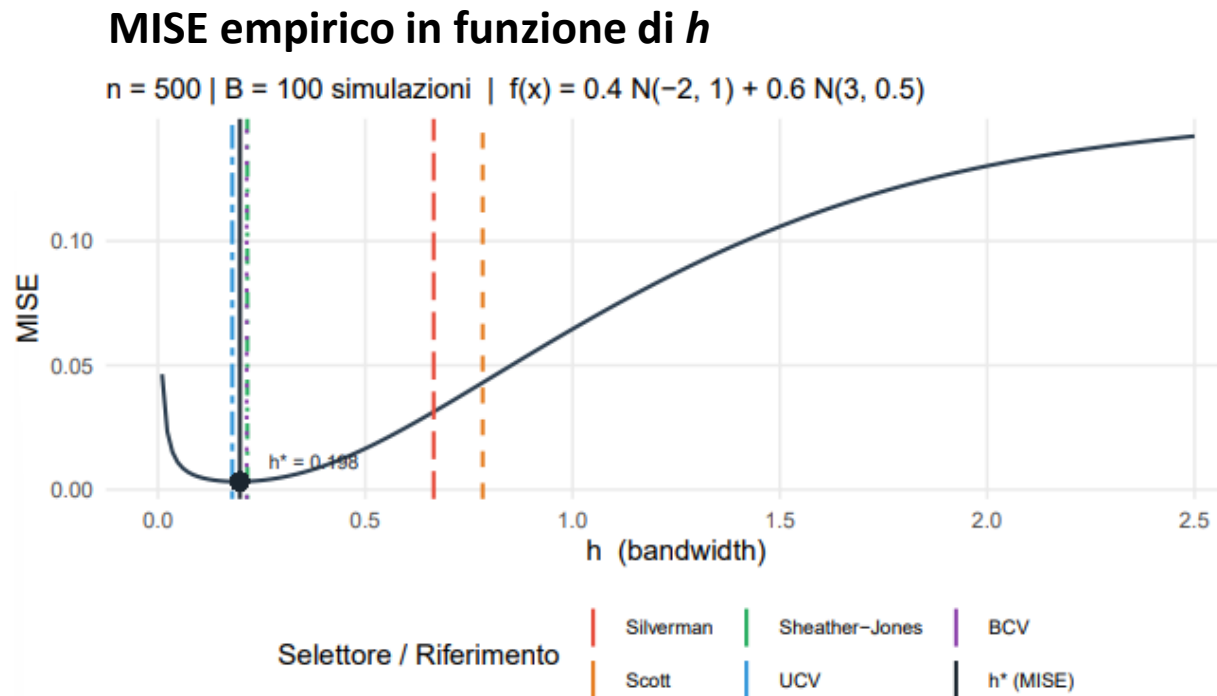
Oversmoothing

$h = 2.00 \rightarrow$ underfitting



Ulteriori casi di studio

Metodo	h	MISE
h^*	0.1977	0.003214
UCV	0.1793	0.003245
BCV	0.2142	0.003292
Sheather-Jones	0.2156	0.003302
Silverman	0.6650	0.031295
Scott	0.7832	0.043037



References

- A Tutorial on Kernel Density Estimation and Recent Advances (Yen-Chi Chen)
- Kernel Density Estimator explained step by step (Medium.com)
- Claude per aiuto sul codice